

# Variational Divergence Minimization

## Mathematical Foundation for GenAI — Week 1

Quiz questions with worked answers and explanations

Topics: deep generative models,  $f$ -divergences, conjugate duality, the  $f$ -GAN min-max objective, reparameterization

### Background in one page

**The goal.** We have data  $D = \{x_1, \dots, x_n\}$  drawn i.i.d. from an unknown distribution  $P_X$ , with each  $x_i \in \mathbb{R}^d$ . We want to *learn to sample* from  $P_X$ . The plan: assume a parametric family  $P_\theta$  (a model), measure how far  $P_\theta$  is from  $P_X$  using a divergence, and minimize that divergence over  $\theta$ .

**The sampler.** Draw a latent  $z \sim \mathcal{N}(0, I)$  and push it through a neural network  $g_\theta$ . The induced distribution of  $x = g_\theta(z)$  is  $P_\theta$  (the *pushforward* of the latent distribution under  $g_\theta$ ). At the optimum  $P_\theta \approx P_X$ , so  $g_\theta(z)$  becomes an approximate sampler for the data.

**The  $f$ -divergence.** For a convex, left-semicontinuous  $f$  with  $f(1) = 0$ ,

$$D_f(P_X \| P_\theta) = \int p_\theta(x) f\left(\frac{p_X(x)}{p_\theta(x)}\right) dx = \mathbb{E}_{x \sim P_\theta} \left[ f\left(\frac{p_X(x)}{p_\theta(x)}\right) \right].$$

Different choices of  $f$  give different divergences (KL, reverse KL, JS, total variation).

**The variational trick.** We don't know the densities, only samples. Using the convex conjugate  $f^*(t) = \sup_u (tu - f(u))$  and the identity  $f(u) = \sup_t (tu - f^*(t))$ , the divergence becomes a *lower bound expressed purely with expectations*:

$$D_f(P_X \| P_\theta) \geq \sup_{T \in \mathcal{T}} \left( \mathbb{E}_{x \sim P_X} [T(x)] - \mathbb{E}_{x \sim P_\theta} [f^*(T(x))] \right).$$

Both expectations can be estimated from samples (law of large numbers). Combined with the generator we get the  $f$ -GAN min-max problem

$$\min_{\theta} \max_{T \in \mathcal{T}} \left( \mathbb{E}_{x \sim P_X} [T(x)] - \mathbb{E}_{x \sim P_\theta} [f^*(T(x))] \right),$$

where  $T$  is the discriminator/critic and  $g_\theta$  is the generator.

**Question 1.** The  $f$ -divergence between two distributions ( $p_X, p_\theta$  are the density functions) is defined as:

- A.  $D_f(P_X \| P_\theta) = \int p_\theta(x) f\left(\frac{p_X(x)}{p_\theta(x)}\right) dx$
- B.  $D_f(P_X \| P_\theta) = \int p_X(x) f\left(\frac{p_X(x)}{p_\theta(x)}\right) dx$
- C.  $D_f(P_X \| P_\theta) = \int p_\theta(x) f\left(\frac{p_\theta(x)}{p_X(x)}\right) dx$
- D.  $D_f(P_X \| P_\theta) = \int p_X(x) f\left(\frac{p_\theta(x)}{p_X(x)}\right) dx$

**Answer: A**

The  $f$ -divergence weights by the *second* distribution  $p_\theta$  (the one we measure divergence *to*) and applies  $f$  to the ratio  $\frac{p_X}{p_\theta}$ . A quick sanity check uses the KL generator  $f(u) = u \log u$ :

$$\int p_\theta \frac{p_X}{p_\theta} \log \frac{p_X}{p_\theta} dx = \int p_X \log \frac{p_X}{p_\theta} dx = D_{\text{KL}}(P_X \| P_\theta),$$

recovering the forward KL exactly. Options B and D put the wrong density in front; C and D invert the ratio. Only A reproduces a valid  $f$ -divergence.

**Question 2.** The  $f$  function used in the divergence should be at least:

- A. Left Continuous
- B. Right Continuous
- C. Left Semi Continuous
- D. Right Semi Continuous

**Answer: C — Left Semi Continuous**

The defining conditions on  $f$  are: convex, *left- (lower-) semicontinuous*, and  $f(1) = 0$ . Lower semicontinuity is the technical requirement that makes the convex conjugate  $f^*$  (and the biconjugate  $f^{**} = f$ ) well behaved, which is what the entire variational lower bound relies on. Full continuity is stronger than needed; semicontinuity is the minimal requirement.

**Question 3.** What is the  $f$  function for KL-divergence?

- A.  $\log(u + 1)$
- B.  $u \log(u)$
- C.  $0.5 |u - 1|$
- D.  $(u + 1) \log(u + 1)$

**Answer: B —  $u \log(u)$**

Substituting  $f(u) = u \log u$  into the definition gives  $\int p_\theta \cdot \frac{p_X}{p_\theta} \log \frac{p_X}{p_\theta} dx = \int p_X \log \frac{p_X}{p_\theta} dx$ , the forward KL. For reference, option C,  $0.5 |u - 1|$ , is the generator of the *total variation* distance.

**Question 4.** Is Forward KL the same as Reverse KL?

- A. Yes
- B. No

**Answer: B — No**

Forward KL is  $D_{\text{KL}}(P_X \| P_\theta)$  and reverse KL is  $D_{\text{KL}}(P_\theta \| P_X)$ ; in general  $D_{\text{KL}}(P \| Q) \neq D_{\text{KL}}(Q \| P)$  — KL is *not symmetric*. They also induce different fitting behavior: forward KL is mass-covering / mean-seeking (it penalizes the model placing zero mass where the data has mass), while reverse KL is mode-seeking (it lets the model concentrate on a subset of modes).

**Question 5.** If  $f(u)$  is a convex function, then the conjugate function  $f^*(t)$  is a:

- A. Concave function
- B. Convex function
- C. Quasi-Convex function
- D. Quasi-Concave function

**Answer: B — Convex function**

The convex conjugate  $f^*(t) = \sup_u (tu - f(u))$  is a pointwise supremum of *affine* functions of  $t$  (each fixed  $u$  gives the line  $t \mapsto tu - f(u)$ ). A pointwise supremum of affine/convex functions is always convex — so  $f^*$  is convex regardless of whether the original  $f$  was convex.

**Question 6. If the divergence is the JS-divergence, what is the  $f$  function used?**

- A.  $0.5(u \log(u+1) - (u+1) \log(0.5(u+1)))$
- B.  $0.5(u \log(u) - (u-1) \log(0.5(u-1)))$
- C.  $0.5(u \log(u-1) - (u+1) \log(0.5(u+1)))$
- D.  $0.5(u \log(u) - (u+1) \log(0.5(u+1)))$

**Answer: D**

The JS generator is  $f(u) = \frac{1}{2}(u \log u - (u+1) \log \frac{u+1}{2})$ , and  $0.5(u+1) = \frac{u+1}{2}$ . Only D has both the correct  $u \log(u)$  term and the correct  $(u+1) \log(0.5(u+1))$  term. A has  $u \log(u+1)$  (wrong first term), B uses  $(u-1)$ , and C has  $u \log(u-1)$ .

**Question 7. Is there a function  $f$  such that the resulting divergence can take negative values?**

- A. Maybe for some values
- B. Yes
- C. No

**Answer: C — No**

An  $f$ -divergence is always non-negative for any valid  $f$ . By Jensen's inequality (using convexity of  $f$ ):

$$D_f(P_X \| P_\theta) = \mathbb{E}_{p_\theta} \left[ f\left(\frac{p_X}{p_\theta}\right) \right] \geq f\left(\mathbb{E}_{p_\theta} \left[ \frac{p_X}{p_\theta} \right]\right) = f(1) = 0,$$

since  $\mathbb{E}_{p_\theta} \left[ \frac{p_X}{p_\theta} \right] = \int p_\theta \cdot \frac{p_X}{p_\theta} dx = \int p_X dx = 1$  and  $f(1) = 0$ . So the divergence can never be negative.

**Question 8. Given the variational representation  $D_f(P_X \| P_\theta) = \sup_{T \in \mathcal{T}} (\mathbb{E}_{x \sim P_X} [T(x)] - \mathbb{E}_{x \sim P_\theta} [f^*(T(x))])$ , which statements correctly describe how this leads to a min-max problem? (Select all that apply.)**

- A. The inner supremum over  $T$  corresponds to the discriminator maximizing the difference between real and generated data.
- B. The generator minimizes the divergence by pushing  $P_\theta$  closer to  $P_X$ .

- C. The convex conjugate  $f^*$  ensures the  $f$ -divergence is symmetric and always tractable.
- D. The objective defines a saddle-point problem.
- E. The generator maximizes the divergence by generating more distinguishable samples.

**Answer: A, B, D**

The full objective is  $\min_{\theta} \max_{T \in \mathcal{T}} (\mathbb{E}_{P_X}[T] - \mathbb{E}_{P_{\theta}}[f^*(T)])$ .

**A** — the inner  $\sup_T$  is the discriminator/critic; maximizing this objective amounts to best distinguishing real ( $P_X$ ) from generated ( $P_{\theta}$ ) samples. **B** — the outer  $\min_{\theta}$  trains the generator  $g_{\theta}$  to push  $P_{\theta} \rightarrow P_X$ . **D** — a minmax objective is a saddle-point problem by definition.

**C is false:**  $f^*$  buys variational *tractability*, not symmetry — most  $f$ -divergences (e.g. KL) are asymmetric. **E is false:** it has the roles backwards — the generator *minimizes* the divergence to make samples *less* distinguishable; the discriminator is the one that maximizes.

**Question 9.** Which of the following are benefits of variational divergence minimization? (Select all that apply.)

- A. Allows divergence estimation via samples instead of density functions.
- B. Requires access to exact density ratios between model and data.
- C. Can be implemented using neural networks for both the generator and discriminator.
- D. Provides flexibility to choose different divergence measures.

**Answer: A, C, D**

**A** — the central benefit: the variational form uses only *samples* from  $P_X$  (the dataset) and  $P_{\theta}$  (outputs of  $g_{\theta}$ ), so the densities are never needed. **C** — both  $g_{\theta}$  and the critic  $T$  are neural networks (the  $f$ -GAN architecture). **D** — varying the convex function  $f$  yields different divergences (KL, reverse KL, JS, TV).

**B is false** — and it is the *opposite* of a benefit. The whole point of the method is to *avoid* needing exact density ratios by working with samples and the variational lower bound.

**Question 10.** With conjugate  $f^*(t) = \sup_{u>0}(tu - f(u))$ , in the variational representation of the  $f$ -divergence, what does the optimal  $T(x)$  represent?

- A. An estimate of the likelihood ratio  $\frac{p_X(x)}{p_{\theta}(x)}$ .
- B. The output of a neural network discriminator  $T_{\vartheta}(x)$ .
- C. The true divergence value at each sample point.

**Answer: A**

The inner supremum is tight where  $T^*(x) = f'\left(\frac{p_X(x)}{p_{\theta}(x)}\right)$  — a function of the density ratio. So the optimal variational function *encodes (an estimate of) the likelihood ratio  $\frac{p_X}{p_{\theta}}$* . Option B describes the *parameterization* we optimize over, not what the optimum *represents*; the question asks the latter. (See Q14 for the

precise functional form.)

**Question 11.** You observe a dataset  $D = \{x_i\}_{i=1}^n$  with  $x_i \in \mathbb{R}^d$  and assume  $x_i \stackrel{\text{i.i.d.}}{\sim} p_X$ . Which statement is correct?

- A. The coordinates inside each  $x_i$  are independent.
- B. The samples  $x_i$  are independent draws from the same  $p_X$ , but coordinates within each  $x_i$  may be dependent.
- C.  $p_X(x) = \prod_{j=1}^d p_X(x_j)$  must hold.
- D.  $p_X$  must be Gaussian.

**Answer: B**

“i.i.d.” is a statement about the *samples* (data points): each  $x_i$  is an independent draw from the same distribution  $p_X$ . It says nothing about the internal structure of a single  $x_i$  — the coordinates within one vector can be strongly dependent (e.g. neighboring pixels in an image). A and C wrongly assert coordinate-wise independence; D wrongly assumes a Gaussian form. Neither is implied by i.i.d.

**Question 12.** Let  $z \sim p_z$  and  $x = g_\theta(z)$ . The induced distribution  $p_\theta$  on  $x$  is best described as:

- A.  $p_\theta(x) = p_z(g_\theta(x))$ .
- B.  $p_\theta$  is the pushforward distribution of  $p_z$  under  $g_\theta$ .
- C.  $p_\theta(x) = \int p_z(z) dz$  (a constant).
- D.  $p_\theta(x) = p_X(x)$  by construction.

**Answer: B**

Sampling  $z \sim p_z$  and applying the deterministic map  $x = g_\theta(z)$  produces, by definition, the *pushforward* measure of  $p_z$  under  $g_\theta$  (written  $p_\theta = (g_\theta)_\# p_z$ ). A misapplies the map and omits the Jacobian (a valid change-of-variables density would be  $p_z(g_\theta^{-1}(x)) |\det J|$ , not  $p_z(g_\theta(x))$ ). C is a meaningless constant ( $\int p_z dz = 1$ ). D holds only *at the optimum*, not by construction.

**Question 13.** Using the variational form with  $f(u) = u \log u$  and  $f^*(t) = e^{t-1}$ , the inner objective becomes:

- A.  $\sup_T (\mathbb{E}_{p_X}[T] - \mathbb{E}_{p_\theta}[\log(1 + e^T)])$
- B.  $\sup_T (\mathbb{E}_{p_X}[T] - \mathbb{E}_{p_\theta}[e^{T-1}])$
- C.  $\sup_T (\mathbb{E}_{p_\theta}[T] - \mathbb{E}_{p_X}[e^{T-1}])$
- D.  $\inf_T (\mathbb{E}_{p_X}[T] - \mathbb{E}_{p_\theta}[e^{T-1}])$

**Answer: B**

Substitute  $f^*(T(x)) = e^{T(x)-1}$  into  $\sup_T (\mathbb{E}_{p_X}[T] - \mathbb{E}_{p_\theta}[f^*(T)])$ . The first expectation is over  $p_X$ , the second over  $p_\theta$ , giving option B. (Check:  $f^*(t) = \sup_u (tu - u \log u)$ ; setting the derivative  $t - \log u - 1 =$

0 gives  $u = e^{t-1}$ , hence  $f^*(t) = e^{t-1}$ .) A uses the JS conjugate  $\log(1 + e^T)$ ; C swaps the two measures; D uses inf instead of sup.

**Question 14.** For a differentiable convex  $f$ , the tightness (Fenchel–Young equality) condition implies the optimal  $T^*(x)$  satisfies:

- A.  $T^*(x) = f\left(\frac{p_X(x)}{p_\theta(x)}\right)$
- B.  $T^*(x) = f'\left(\frac{p_X(x)}{p_\theta(x)}\right)$
- C.  $T^*(x) = \left(\frac{p_X(x)}{p_\theta(x)}\right)^{-1}$
- D.  $T^*(x) = \log \frac{p_\theta(x)}{p_X(x)}$

**Answer: B**

The Fenchel–Young inequality  $f(u) + f^*(t) \geq ut$  holds with equality exactly when  $t = f'(u)$ . In the bound the ratio plays the role of  $u = \frac{p_X}{p_\theta}$ , so the optimal critic is  $T^*(x) = f'\left(\frac{p_X(x)}{p_\theta(x)}\right)$ . This refines Q10:  $T^*$  encodes the likelihood ratio (Q10), and its precise form is  $f'$  of that ratio (here). Option A applies  $f$  instead of  $f'$ ; C and D are specific forms that coincide with  $f'$  only for particular  $f$ , not in general.

**Question 15.** Let  $L(\theta) = \mathbb{E}_{z \sim p_z}[\ell(g_\theta(z))]$  with  $z$  independent of  $\theta$ . Which gradient expression is correct?

- A.  $\nabla_\theta L = \mathbb{E}_z[\ell(g_\theta(z)) \nabla_\theta \log p_z(z)]$
- B.  $\nabla_\theta L = \mathbb{E}_z[J_{g_\theta}(z)^\top \nabla_x \ell(x) | x = g_\theta(z)]$
- C.  $\nabla_\theta L = \nabla_\theta \int p_z(z) dz$
- D.  $\nabla_\theta L = \mathbb{E}_{x \sim p_X}[\nabla_\theta \ell(x)]$

**Answer: B**

Because the latent distribution  $p_z$  does not depend on  $\theta$ , the gradient moves inside the expectation and the chain rule gives the Jacobian of  $g_\theta$  times the gradient of  $\ell$ :

$$\nabla_\theta L = \mathbb{E}_z[J_{g_\theta}(z)^\top \nabla_x \ell(x)|_{x=g_\theta(z)}].$$

This is the *pathwise / reparameterization* gradient. A is a score-function (REINFORCE) term, but here  $\nabla_\theta \log p_z(z) = 0$  since  $p_z \perp \theta$ , so it vanishes. C is zero ( $\int p_z dz = 1$ , constant). D differentiates over the data measure  $p_X$ , which has no  $\theta$  dependence.

### Answer key at a glance

|             |        |        |             |
|-------------|--------|--------|-------------|
| Q1: A       | Q2: C  | Q3: B  | Q4: B       |
| Q5: B       | Q6: D  | Q7: C  | Q8: A, B, D |
| Q9: A, C, D | Q10: A | Q11: B | Q12: B      |
| Q13: B      | Q14: B | Q15: B |             |

**A note on Q10 vs. Q14**

These two can look contradictory but are not. Q10 asks, among coarse options, what the optimal  $T^*$  *represents* — the answer is “the likelihood ratio” (A). Q14 asks for the *precise* functional form under Fenchel–Young tightness — the answer is  $f'$  of that ratio (B). The optimal critic is  $T^*(x) = f'\left(\frac{p_X}{p_\theta}\right)$ ; it is a monotone transform of the density ratio, so it “encodes” the ratio without being equal to it for a general  $f$ .